CSCI 5380 - Network Virtualization and Orchestration

Lab 9
Automate VM, VN, Docker, and BGP path

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Summary:

In this lab, you will use what you have learned in previous labs and automate the processes into a single application.

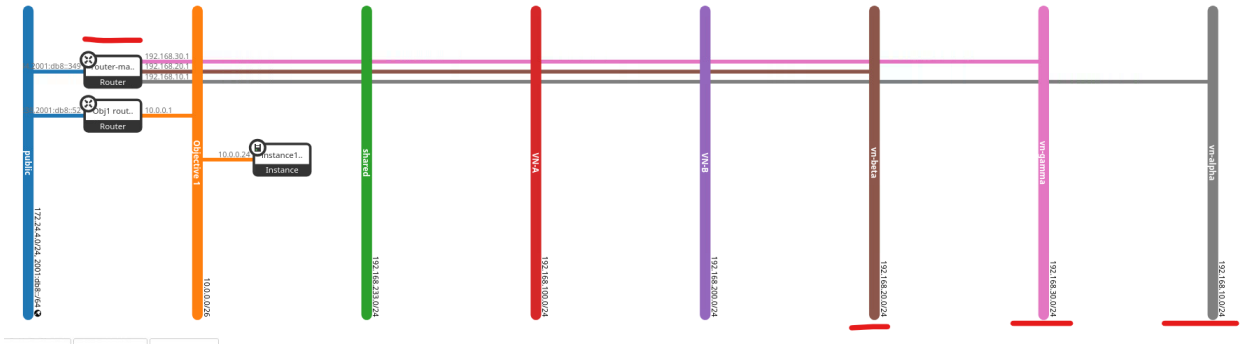
Required technologies:

- BGP
- Hypervisor/Orchestrator (such as OpenStack)
- Containers
- SDN Controller
- Hardware server
- Service-chain

Objectives: Virtualized Network Automation

Create an application that meets the following functionality (each objective must be a separate Python module in your code i.e. your main .py file should import the different modules you write):

1) Automate the creation of multiple virtual networks (VNs) within the hypervisor and their connection to the public network.



Python module: create_VMs.py automatically creates 3 Virtual Networks: vn-alpha, vn-gamma and vn-beta and assigns them a subnet and it also creates a common gateway router (router-main) and connects all these VNs.

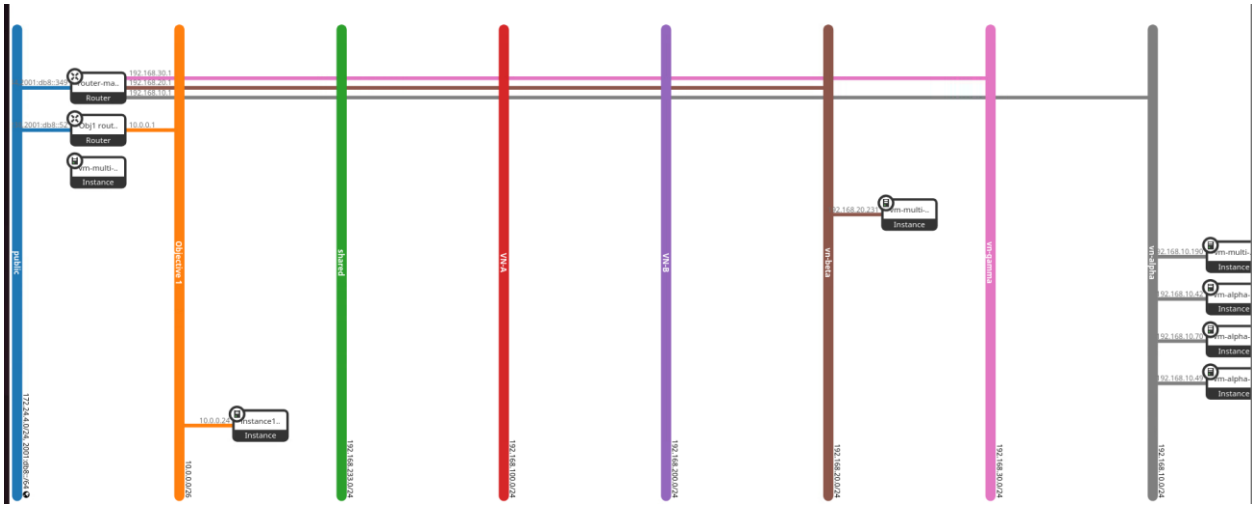
2) Automate the creation of multiple VMs within the hypervisor-

a) Both single tenant (same VN) and multi-tenant (different VNs).

b) All VMs should be accessible from the host server and be able to access the Internet.

create_VMs.py will create single and multi tenant VMs, assign them to a VN and then also assign them a floating IP.

IP connectivity shown in next objectives



Floating IPs

Displaying 19 items

Project	IP Address	Description	Mapped Fixed IP Address	Pool	Status	DNS Name	DNS Domain	Actions
admin	172.24.4.222		-	public	Down			Release Floating IP
admin	172.24.4.216	Floating IP2	-	public	Down			Release Floating IP
admin	172.24.4.31		-	public	Down			Release Floating IP
admin	172.24.4.192		vm-multi-a 192.168.10.190	public	Active			Disassociate
Lab2	172.24.4.198		VN-A_VM-1 192.168.100.245	public	Active			Disassociate
admin	172.24.4.170		-	public	Down			Release Floating IP
admin	172.24.4.53		vm-alpha-02 192.168.10.70	public	Active			Disassociate
admin	172.24.4.213		vm-multi-b 192.168.20.231	public	Active			Disassociate
admin	172.24.4.6		-	public	Down			Release Floating IP
admin	172.24.4.154		vm-alpha-01 192.168.10.49	public	Active			Disassociate
admin	172.24.4.205	Floating IP1	instance1-1 10.0.0.24	public	Active			Disassociate
admin	172.24.4.124		-	public	Down			Release Floating IP
lab_3_service_chaining	172.24.4.112		-	public	Down			Release Floating IP

3) Automate the security groups and port security configuration to make intra-VN and inter-VN communication possible.

Security Groups

Filter | + Create Security Group | Delete Security Groups

Displaying 6 items

Name	Security Group ID	Description	Shared	Actions
allow-all	2cbd8dc9-278b-4be1-8962-bb4c58a4d37c	Allow all — refined by module 3	-	Manage Rules
allowall	80ae54cb-4699-490d-bc58-eb84442ae803		-	Manage Rules
default	81b93943-7c8f-483a-b99a-4341cf1300df	Default security group	-	Manage Rules
sg-host-access	65905b4c-2275-44fd-84b1-f69b4d684f80	Host access: SSH (22) + ICMP from 0.0.0.0/0	-	Manage Rules
sg-inter-vn	fe68beba-73a1-4931-9a90-5ab89b9ff249	Inter-VN: allow all traffic between VMs on different networks	-	Manage Rules
sg-intra-vn	eb77c5ce-6ac1-414c-8eed-cfbbbe6b73e9	Intra-VN: allow all traffic between VMs on the same network	-	Manage Rules

Displaying 6 items

Manage Security Group Rules: sg-host-access (65905b4c-2275-44fd-84b1-f69b4d684f80)

+ Add Rule | Delete Rule

Displaying 4 items

Direction	Ether Type	IP Protocol	Port Range	Remote IP Prefix	Remote Security Group	Description	Actions
Egress	IPv4	Any	Any	0.0.0.0/0	-	-	Delete Rule
Egress	IPv6	Any	Any	::/0	-	-	Delete Rule
Ingress	IPv4	ICMP	Any	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	22 (SSH)	0.0.0.0/0	-	-	Delete Rule

Displaying 4 items

```
login as 'cirros' user. default password: 'gocubsgo'. use 'sudo' for root.
vm-multi-b login: cirros
Password:
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1442 qdisc pfifo_fast qlen 1000
    link/ether fa:16:3e:bd:5a:c4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.231/24 brd 192.168.20.255 scope global dynamic noprefixroute eth0
       valid_lft 37755sec preferred_lft 32355sec
    inet6 fe80::f816:3eff:febd:5ac4/64 scope link
       valid_lft forever preferred_lft forever
$ ping 192.168.10.42
PING 192.168.10.42 (192.168.10.42) 56(84) bytes of data.
64 bytes from 192.168.10.42: icmp_seq=1 ttl=63 time=7.14 ms
64 bytes from 192.168.10.42: icmp_seq=2 ttl=63 time=1.72 ms
64 bytes from 192.168.10.42: icmp_seq=3 ttl=63 time=0.507 ms
64 bytes from 192.168.10.42: icmp_seq=4 ttl=63 time=1.76 ms
^C
--- 192.168.10.42 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3016ms
rtt min/avg/max/mdev = 0.507/2.780/7.142/2.567 ms
$
```

Internet connectivity

```
rtt min/avg/max/mdev = 0.507/2.780/7.142/2.567 ms
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=6.30 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=111 time=3.01 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=111 time=3.61 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=111 time=5.90 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=111 time=3.82 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 3.008/4.527/6.297/1.316 ms
$
```

4) Automate spinning up and configuring a Quagga/FRR BGP router as a Docker container.
    a) Automate its BGP configuration to peer with the SDN controller in the next objective.

Objective 4 and 5 done together

5) Automate spinning up and configuring an SDN controller as another Docker container.
    a) Automate its BGP speaker configuration to peer with Quagga/FRR.

For objective 4 and 5, I've used FRR and Ryu as the SDN controller. To orchestrate the containers and the network settings, I have used docker-compose. The File system for this look like this:

```
project/
├── docker-compose.yml
├── frr_config/
│   ├── frr.conf      ← BGP config, peers, advertised prefixes
│   ├── daemons       ← enables zebra + bgpd only
│   └── vtysh.conf
└── ryu_build/
    ├── Dockerfile     ← builds the Ryu image
    ├── ryu_bgp_app.py ← OpenFlow switch + BGP speaker app
    └── ryu-bgp.conf   ← Ryu manager settings
```

(check individual files and how they work. Each of them have detailed descriptions of their working)

```
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$ docker compose up -d
 [+] up 0/1
  ⠿ Image ryu-bgp-speaker:latest Pulling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  ⠿ 0.5s
 [+] Building 35.7s (15/15) FINISHED
  => [internal] load local bake definitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  => => transferring context: 2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  => => transferring context: 5.22kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  => [1/8] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f                                                                                                                                                                                                                                                                                                                                                
  => => resolve docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f                                                                                                                                                                                                                                                
  => => sha256:a3f1dfe736c5f959143f23d75ab522a60be2da902efac236f4fb2a153cc14a5d 14.53MB / 14.53MB                                                                                                                                                                                                                                                                
  => => sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f 10.41kB / 10.41kB                                                                                                                                                                                                                                                                
  => => sha256:314bc2fb0714b7807bf5699c98f0c73817e579799f2d91567ab7e9510f5601a5 1.75kB / 1.75kB                                                                                                                                                                                                                                                                
  => => sha256:b5f62925bd0f63f48cc8acd5e87d0c3a07e2f229cd2fb0a9586e68ed17f45ee3 5.25kB / 5.25kB                                                                                                                                                                                                                                                                
  => => sha256:302e3ee498053a7b5332ac79e8efebec16e900289fc1ecd1c754ce8fa047fcab 29.13MB / 29.13MB                                                                                                                                                                                                                                                                
  => => sha256:030d7bdc20a63e3d22192b292d006a69fa3333949f536d62865d1bd0506685cc 3.51MB / 3.51MB                                                                                                                                                                                                                                                                
  => => sha256:3971691a363796c39467aae4cdce6ef773273fe6bfc67154d01e1b589befb912 248B / 248B                                                                                                                                                                                                                                                                
  => => extracting sha256:302e3ee498053a7b5332ac79e8efebec16e900289fc1ecd1c754ce8fa047fcab                                                                                                                                                                                                                                                                
  => => extracting sha256:030d7bdc20a63e3d22192b292d006a69fa3333949f536d62865d1bd0506685cc                                                                                                                                                                                                                                                                
  => => extracting sha256:a3f1dfe736c5f959143f23d75ab522a60be2da902efac236f4fb2a153cc14a5d                                                                                                                                                                                                                                                                
```

```
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$ docker compose up -d
WARN[0000] No services to build
[+] up 2/2
 ✔ Container frr-bgp-router      Created
 ✔ Container ryu-sdn-controller Created
```

BGP neighborship verification:

```
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$ docker exec -it frr-bgp-router vtysh -c "show bgp summary"

IPv4 Unicast Summary (VRF default):
BGP router identifier 172.20.0.2, local AS number 65001 vrf-id 0
BGP table version 0
RIB entries 5, using 960 bytes of memory
Peers 1, using 717 KiB of memory

Neighbor        V         AS   MsgRcvd   MsgSent   TblVer  InQ OutQ  Up/Down State/PfxRcd   PfxSnt Desc
172.20.0.3      4      65002         6         6        0    0    0 00:00:29            0        0 Ryu-SDN-Controller

Total number of neighbors 1
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$
```

```
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$ docker logs ryu-sdn-controller | grep -i bgp
::'), 'bgp_server_port': 179, 'refresh_stalepath_time': 0, 'refresh_max_eor_time': 0, 'label_range': (100, 100000), 'allow_local_as_in_count': 0, 'cluster_id': None, 'local_pre
f': 100}
API method neighbor.create called with args: {'ip_address': '172.20.0.2', 'remote_as': 65001, 'remote_port': 179, 'peer_next_hop': None, 'password': None, 'is_route_server_clie
nt': False, 'is_route_reflector_client': False, 'is_next_hop_self': False, 'connect_mode': 'active', 'cap_enhanced_refresh': False, 'cap_four_octet_as_number': True, 'cap_mbgp_
ipv4': True, 'cap_mbgp_ipv6': False, 'cap_mbgp_vpnv4': False, 'cap_mbgp_vpnv6': False, 'cap_mbgp_evpn': False, 'cap_mbgp_ipv4fs': False, 'cap_mbgp_ipv6fs': False, 'cap_mbgp_vpn
v4fs': False, 'cap_mbgp_vpnv6fs': False, 'cap_mbgp_l2vpnfs': False}
BGP neighbor added: 172.20.0.2 AS 65001
instantiating app ryu.controller.ofp_handler of OFPHandler
Sent notification to ('172.20.0.3', '179') >> BGPNotification(data=b'',error_code=6,error_subcode=5,len=21,type=3)
Connection to peer: 172.20.0.2 established
Negotiated hold time (30) is lower then configured/default (40).
BGP PEER UP    172.20.0.2  AS 65001
loading app /ryu-app/ryu_bgp_app.py
instantiating app /ryu-app/ryu_bgp_app.py of BGPSwitchApp
Starting BGP speaker AS=65002 router-id=172.20.0.3
API method core.start called with args: {'waiter': <ryu.lib.hub.Event object at 0x7ad510eb1d00>, 'local_as': 65002, 'router_id': '172.20.0.3', 'bgp_server_hosts': ('0.0.0.0', '
::'), 'bgp_server_port': 179, 'refresh_stalepath_time': 0, 'refresh_max_eor_time': 0, 'label_range': (100, 100000), 'allow_local_as_in_count': 0, 'cluster_id': None, 'local_pre
f': 100}
API method neighbor.create called with args: {'ip_address': '172.20.0.2', 'remote_as': 65001, 'remote_port': 179, 'peer_next_hop': None, 'password': None, 'is_route_server_clie
nt': False, 'is_route_reflector_client': False, 'is_next_hop_self': False, 'connect_mode': 'active', 'cap_enhanced_refresh': False, 'cap_four_octet_as_number': True, 'cap_mbgp_
ipv4': True, 'cap_mbgp_ipv6': False, 'cap_mbgp_vpnv4': False, 'cap_mbgp_vpnv6': False, 'cap_mbgp_evpn': False, 'cap_mbgp_ipv4fs': False, 'cap_mbgp_ipv6fs': False, 'cap_mbgp_vpn
v4fs': False, 'cap_mbgp_vpnv6fs': False, 'cap_mbgp_l2vpnfs': False}
BGP neighbor added: 172.20.0.2 AS 65001
Sent notification to ('172.20.0.3', '179') >> BGPNotification(data=b'',error_code=6,error_subcode=5,len=21,type=3)
BGP PEER UP    172.20.0.2  AS 65001
(.venv) student@csci5380-vm1-prpr4002:~/nvo_lab/Openstack automation$
```

Deliverable:

Create a personal GitHub page that demonstrates the required functionality.